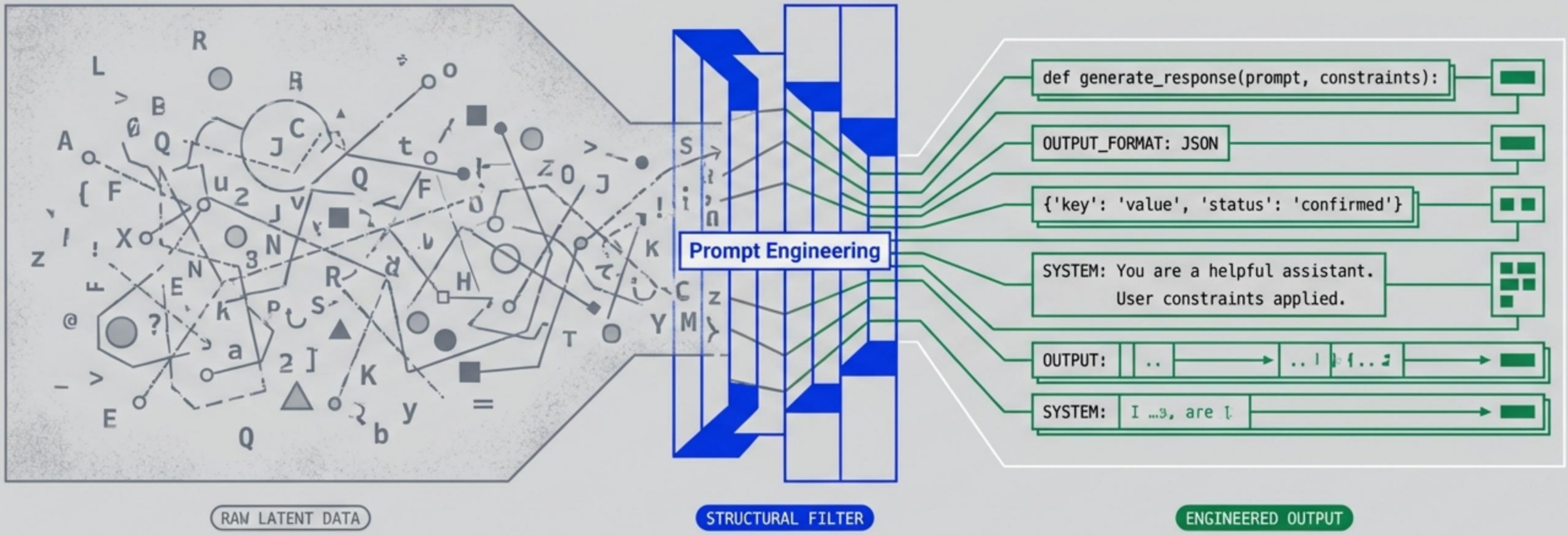
The Prompt Engineering Toolkit

17 Techniques to Master Large Language Models (From 1B to Frontier Models)

NotebookLM

Taming the Latent Space

Even smaller 1B-parameter models possess immense underlying capability. However, extracting accurate, creative, or reliable responses requires imposing structure on chaos. Prompt engineering is not merely asking questions—it is programming the model's behaviour using natural language constraints.



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1. Zero-Shot Prompting

Ask the model to perform a task or answer a question directly, without any examples.

Simple Q&A | Brainstorming | Basic Summaries

```
[BEFORE] What is photosynthesis?
```

```
[AFTER] Explain photosynthesis to a
5-year-old child.
Engineeers model photosynthess to
send it.
```

2. Few-Shot Prompting

Provide input/output examples to demonstrate the desired pattern and format.

Sentiment Analysis | Specific Formats | Data Extraction

```
[BEFORE] Classify the sentiment of
this movie review: "The movie was
okay, not great but not terrible."
```

```
[AFTER] Review: "Loved it!"
Sentiment: Positive.
Review: "The movie was okay..."
Sentiment:
```

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3. Role Prompting

Instruct the LLM to adopt a specific character or expert persona.

Engaging Dialogues | Expert Perspectives | Niche Audiences

```
[BEFORE] Tell me about black holes.
```

```
[AFTER] You are Professor Astra, a friendly astronomer. Tell me about black holes!
```

4. Style Prompting

Guide the LLM to write in a specific literary, formal, or artistic style.

Creative Writing | Brand Voice | Content Adaptation

```
[BEFORE] Write a short description of a sunset.
```

```
[AFTER] Write a short description of a sunset in the style of a haiku.
```

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5. Emotion Prompting

Generate responses conveying a specific emotion or from an emotional perspective.

Empathetic Support | Personal Notes | Emotional Depth

```
[BEFORE] Write a thank you note for
a gift I received.
```

```
[AFTER] Write a thank you note for
a gift I received. Make it sound
very excited and deeply
grateful...
```

6. Contextual Prompting

Provide the LLM with sufficient background history, preferences, or constraints.

Personalised Recs | Problem Solving | Multi-turn Chat

```
[BEFORE] Suggest a gift.
```

```
[AFTER] I need a gift suggestion.
Recipient: My sister, 30.
Occasion: Birthday.
Interests: Fantasy novels,
gardening, tea. Budget: Around $50.
```

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7. Chain-of-Thought (CoT)

Force the LLM to think step by step to expose its reasoning and improve accuracy.

Maths Problems | Logic Puzzles | Debugging

```
[BEFORE] Roger has 5 tennis balls. He buys 2
more cans of tennis balls. Each can has 3 tennis
balls. How many tennis balls does he have now?
```

```
[AFTER] Roger has 5 tennis balls. He buys 2 more
cans of tennis balls. Each can has 3 tennis
balls. How many tennis balls does he have now?
Let's think step by step.
```

8. System Prompting

Provide high-level, persistent rules or persona guidelines across an entire session.

Consistent Behaviour | Standing Instructions | Guardrails

```
[BEFORE] Summarise the following text in one
sentence: '[text]'
```

```
[AFTER]
System: You are a 'Concise Summariser'. Your
primary goal is to provide the shortest possible
summary, regardless of the topic.
User: [text_to_summarise]
```

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9. Explicit Instructions

Be direct and unambiguous regarding format or content, leaving zero room for misinterpretation.

Strict Formatting | Avoiding Topics | Conciseness

```
[BEFORE] Write about apples.
```

```
[AFTER] Write a short paragraph about
apples focusing on nutritional benefits.
Exactly 3 sentences long. Mention 2
varieties. Do not discuss history.
varieties. Do not discuss history.
```

10. Output Priming

Nudge the model by providing the exact opening words of its desired response.

Forcing JSON/Lists | Setting Tone Instantly

```
[BEFORE] What are the ingredients for a
simple vanilla cake?
```

```
[AFTER] What are the ingredients? Please
list them out.
Here are the ingredients:
-
```

Hooks the next token

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User Input → Model Reflects → Final Output

11. Rephrase and Respond (RaR)

Ask the LLM to explain its understanding of the task before generating the main response.

Ambiguous Requests | High-Stakes Tasks

```
[BEFORE] Write a story about a journey.
```

```
[AFTER] First, briefly describe what kind of
journey you are planning to write about...

Then, write the story.
```

12. Step-Back Prompting

Extract broader principles or definitions before addressing a highly specific query.

Nuanced Distinctions | First-Principles Thinking

```
[BEFORE] Is a tomato a fruit or a vegetable?
```

```
[AFTER] First explain:
1. What is the botanical definition of a fruit?
2. What is the culinary understanding of a
vegetable?
Now, classify a tomato.
```

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13. Self-Critique & Refinement

Instruct the LLM to generate, evaluate against criteria, and revise its own output.

Creative Refining | Code Review | Complex Explanations

```
[BEFORE]
Write a short, catchy slogan for a new
eco-friendly water bottle.
```

```
[AFTER]
1. Generate one initial slogan.
2. Critically evaluate your slogan…
3. Based on your critique, provide an improved
   slogan.
```

14. Goal Decomposition

Break down a massive task into sequential, manageable **sub-goals** within the prompt.

Event Planning | Long Reports | Multi-part Workflows

```
[BEFORE]
Plan a simple weekend trip to a nearby nature
spot.
```

```
[AFTER]
Create a plan that includes:
(1) Suggest a specific spot.
(2) List essential items.
(3) Suggest one main activity.
(4) Provide a safety tip.
```

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15. Meta-Prompting

Use the LLM to write a significantly better, highly structured prompt for itself.

Prompt Optimisation | Blank Page Syndrome

```
[BEFORE]
Give me some fantasy story ideas.
```

```
[AFTER]
I want to use an LLM to generate 3
distinct fantasy ideas. Please write out
the actual, detailed prompt I should use.
```

16. ReAct (Reason + Act)

Interleave reasoning (breaking down problems) with **acting** (simulating lookups).

Fact Verification | Multi-Hop Q&A | Agentic Tasks

```
[BEFORE]
Who was the U.S. president when the first
person walked on the moon?
```

```
[AFTER]
To answer this, follow a ReAct process.
For each step, state:
Thought: [...]
Action: [...]
Observation: [...]
Then, provide the final answer.
```

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17. Thread-of-Thought (ThoT)

Force the LLM to maintain a coherent, connected line of reasoning across multiple turns or vast text generations.

Long-Form Articles | Complex Essays | Extended Problem Solving

```
[BEFORE]
Explain how a bill becomes a law in the US.
```

```
[AFTER]
Please structure your explanation as follows:
1. Introduction...
2. Bill Introduction...
Throughout your explanation, ensure each stage logically follows the previous one, maintaining a
clear thread.
```

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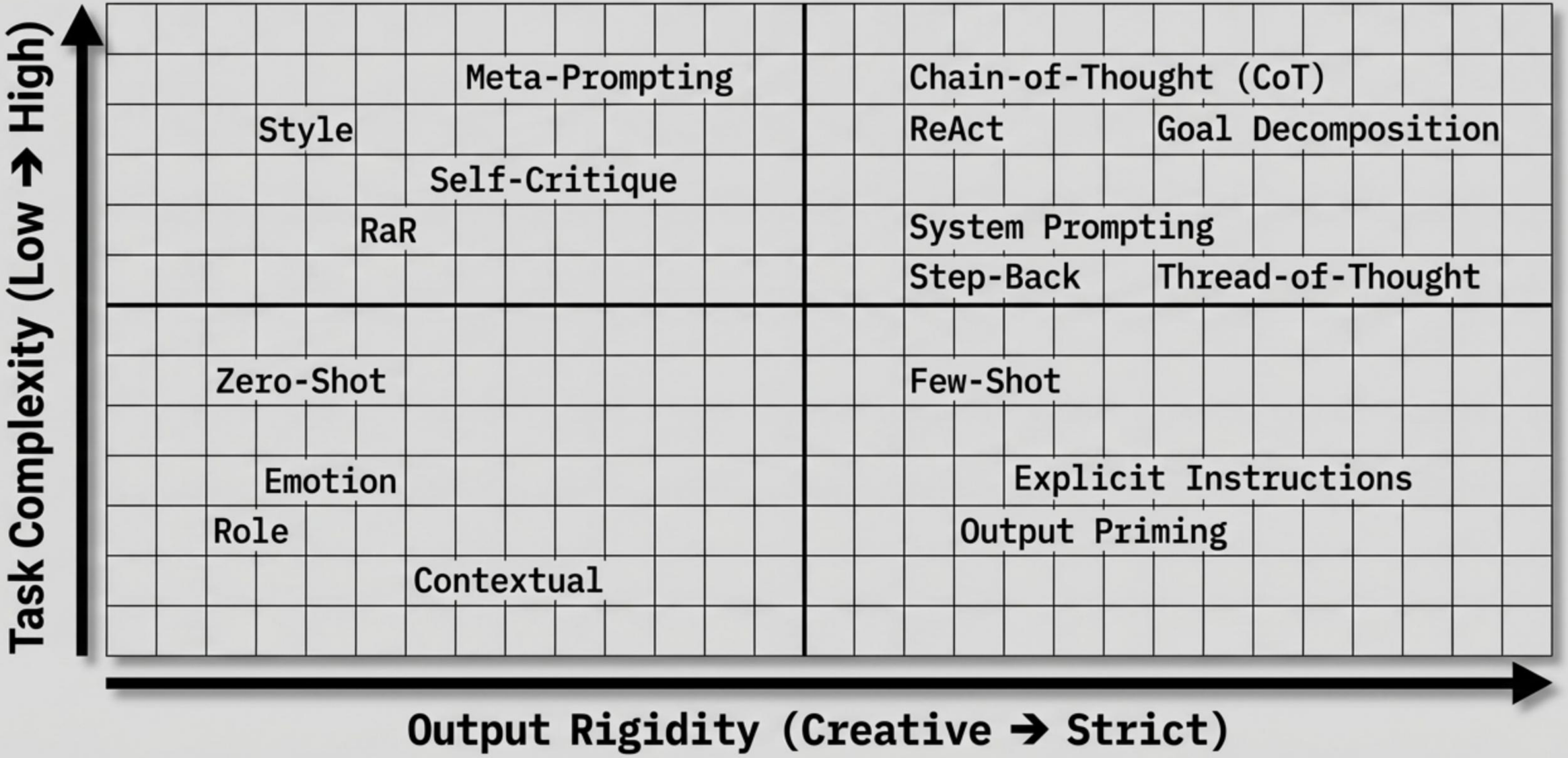
The Prompt Engineer's Diagnostic Matrix



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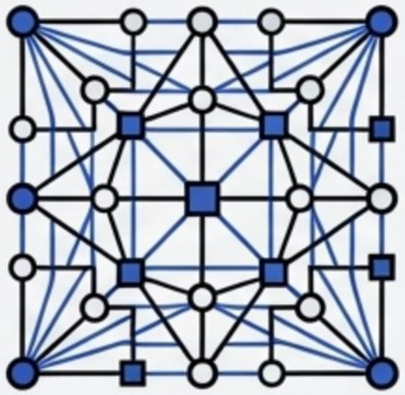
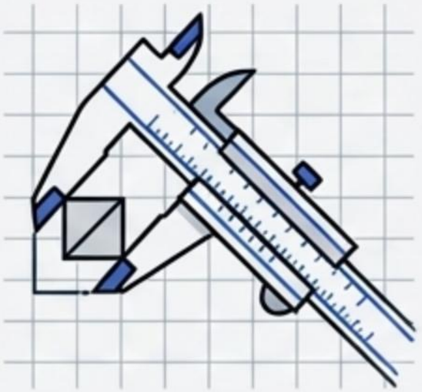
Engineering Better Outputs



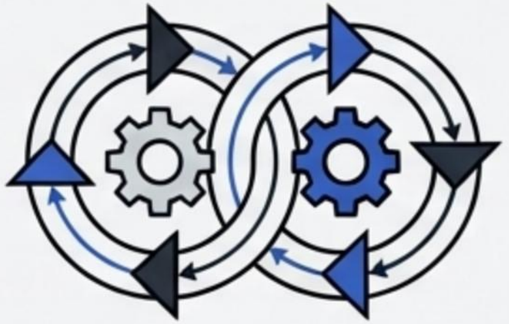
Context is King

Models cannot read minds. The more constraints, personas, and background context you provide provide (Contextual, System, Explicit), the less the LLM hallucinates.



Force the Cognition

Do not let the model jump blindly to conclusions. Use CoT, Step-Back, and ReAct to expose, structure, and enforce its step-by-step thinking process.



Iterate and Scaffold

Treat prompting as a multi-step workflow. Use Meta-Prompting to draft the request, Goal Decomposition to build the structure, and Self-Critique to refine the output.

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